Aayush Alpesh Desai

Curriculum Vitae

Schegargasse 6/1
1190 Vienna, Austria
✉ *desaiaayush08@gmail.com*
ORCID: 0009-0008-9877-5512

Profile

PhD researcher in astrophysics at the Institute of Science and Technology Austria (ISTA), studying magnetic white dwarfs and compact stellar remnants. Research combines optical, ultraviolet, X-ray, and time-domain observations with statistical analysis and numerical modelling to investigate stellar evolution and accretion phenomena.

Education

2023–Present **PhD in Astrophysics**, *Institute of Science and Technology Austria (ISTA)*, Austria.

2020–2023 **Bachelor of Science in Physics**, *Lund University*, Sweden.

Research Experience

Sep. 2023–Present **PhD Researcher**, *Institute of Science and Technology Austria (ISTA)*, Austria.
- Study magnetic white dwarfs, with an emphasis on rapidly rotating merger remnants.
- Analyse optical, ultraviolet, and X-ray observations together with numerical modelling to investigate magnetic white dwarfs and compact stellar merger remnants.

Apr. 2023–Aug. 2023 **Research Intern in Asteroseismology**, *Institute of Science and Technology Austria (ISTA)*, Austria.
- Estimated global asteroseismic parameters using power-spectral-density inversions in a Bayesian framework.

Sep. 2022–Jan. 2023 **Bachelor's Thesis: Galactic Motion of the Sun**, *Lund University*, Sweden.
- Measured the circular velocity of the Sun using Gaia Data Release 3.

Feb. 2022–Jun. 2023 **Deep Learning for High-Energy Physics**, *Lund University*, Sweden.
- Applied deep neural networks to experimental simulations for energy reconstruction.

Feb. 2022–Jun. 2022 **Quantum Emitters in Optical Microcavities**, *Leiden University*, Netherlands.
- Characterised and controlled micrometre-scale optical microcavities and studied single-emitter light–matter interactions.

Jun. 2021–Sep. 2021 **Photoionisation of the Carbon II Ion**, *Lund University*, Sweden.
- Studied Carbon II photoionisation using the R-matrix method.
- Results are documented in a technical report.

Jun. 2018–Jun. 2019 **Beamline for Schools (BL4S)**, *CERN*, Switzerland.
- Won CERN's Beamline for Schools competition as part of the Cryptic Ontics team.
- Conducted the proposed experiment at CERN; results were published in *The Physics Educator*.

Invited Talks

31 Jan. 2025 **Pizza Seminar**, *University of Vienna*, Austria.
"Trust but Verify: Gradients"

Peer-Reviewed Publications

2026 **A. Desai** et al., "Circumstellar interaction in the extreme white dwarf merger remnant ZTF J1901+1458: A new class of white dwarf merger remnants with X-ray emission," *Astronomy & Astrophysics*, 712, A230. doi:10.1051/0004-6361/202557035.

2026 A. Torralba et al. (including **A. Desai**), "A Black Hole Star at Cosmic Noon: Extreme Balmer Break, Photospheric Continuum, and Broad Absorption by Thick Winds in a Little Red Dot at $z = 1.7$," *The Astrophysical Journal Letters*, 1005(2), L37. doi:10.3847/2041-8213/ae7bfd.

2026 B. R. B. Liagre, **A. Desai**, L. Einramhof, and L. Bugnet, "Near-degeneracy effects in quadrupolar mixed modes: From an asymptotic description to data fitting," *Astronomy & Astrophysics*, 707, A321. doi:10.1051/0004-6361/202558023.

2026 A. A. Cristea et al. (including **A. Desai**), "A half ring of ionized circumstellar material trapped in the magnetosphere of a white dwarf merger remnant: A new class of white dwarf merger remnants with X-ray emission," *Astronomy & Astrophysics*, 706, A188. doi:10.1051/0004-6361/202556432.

2025 M. Zu, **A. Desai**, and C. P. Goodrich, "Fully Independent Response in Disordered Solids," *Physical Review Letters*, 134(23), 238201. doi:10.1103/PhysRevLett.134.238201.

2020 S. Chatterji et al. (including **A. Desai**), "A Highschooler's Guide to GeV-Range Electromagnetism," *The Physics Educator*, 2(3), 2050013. doi:10.1142/S2661339520500134.

A complete publication list is attached as a separate document.

Selected Achievements and Awards

Dec. 2021 **Erasmus+ Traineeship Grant**, *Erasmus+*.
Awarded for a research internship at Leiden University.

Oct. 2018 **Recognition from the Department of Atomic Energy**, *Government of India*.
Felicitated by K. N. Vyas, Chairman of the Bhabha Atomic Research Centre, for representing India at CERN's Beamline for Schools competition.

Jun. 2017 **Sakura Science Program**, *Japan Science and Technology Agency (JST)*.
Represented India during a 15-day scientific exchange visit to Japan.

Technical Skills

Programming	Python (advanced); C/C++ and Fortran (intermediate)
Tools	Git, LaTeX, Anaconda, Numba, GitHub, and Microsoft Office
Languages	English (C2), Hindi, and Gujarati